

The client can call any of these three methods to get the elapsed time associated with the timer, in different formats:

- `getTime()` ❸ returns the elapsed time as a floating-point number.
- `getTimeInSeconds()` ❹ returns the elapsed time as an integer number of seconds.
- `getTimeInHHMMSS()` ❺ returns the elapsed time as a formatted string.

The `getTime()` method calls `time.time()` to get the current time and subtracts the starting time to get the elapsed time. The other two methods each make a call to the `getTime()` method of this class to calculate the elapsed time, then do different processing on the output: `getTimeInSeconds()` converts the time into an integer number of seconds, and `getTimeInHHMMSS()` formats the time into a string in *hours:minutes:seconds* format. The output of each of these methods is intended to be sent to a `DisplayText` object (defined in the `pygwidgets` package) to be shown in the window.

The `stop()` method ❻ can be called to stop the timer (for example, when the user completes the puzzle).

The main file for this version of the Slider Puzzle game is available with the rest of the book's resources, at *SliderPuzzles/Main_SliderPuzzleCountUp.py*. It instantiates a `CountUpTimer` object before the main loop begins and saves it in the variable `oCountUpTimer`. It then calls the `start()` method right away. It also creates a `DisplayText` field to display the time. Each time through the main loop, the main code calls the `getTimeInHHMMSS()` method and shows the result in the field:

```
timeToShow = oCountUpTimer.getTimeInHHMMSS() # ask the Timer object for the
elapsed time
oTimerDisplay.setValue('Time: ' + timeToShow) # put that into a text field
```

The variable `oTimerDisplay` is an instance of the `pygwidgets.DisplayText` class. The `setValue()` method of the `DisplayText` class is optimized to check whether the new text to be displayed is the same as the previous text. Therefore, even though we are telling the field to display the amount of time 30 times every second, there is not much work done until the time changes, once per second.